

Practical 6 - Competing risks 1

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `aaatrial.RData`

R packages required: `survival`, `cmprsk`

```
library(survival)
library(cmprsk)
```

Introduction

In this practical we will use a dataset from a randomised trial of screening for abdominal aortic aneurysms (AAA), in which older men were randomised to be invited to be screened for AAA (invited group) or not (control group). The main outcome of interest was deaths relating to AAA (including deaths following AAA repair operations), but men in the study also died from other causes. There are 13,562 people in the data.

If an AAA was found at screening, a repair operation was carried out, essentially removing the risk of future death from AAA. However, the operation itself carries around a 5% risk of mortality. Undetected AAAs may rupture at any time, resulting in either an emergency repair operation (which carries around a 40% risk of mortality) or death. Participants were recruited to the trial between 1997 and 1999 and followed up until the end of 2007.

The variables used in this practical are described below.

Variable	Description
id	Unique identifier for each participant
group	Randomisation group: 0=Control, 1=Invited
dateran	Date of randomisation

Variable	Description
aaadeath	AAA-death indicator: 0=Censored or non-AAA-death, 1=AAA-death
allddeath	All-cause mortality: 0=Censored, 1=Death (any cause)
deathtype	0=Censored; 1=non-AAA death; 2=AAA-death
timeout	Date of study exit

This is already an R dataset, and the dates have been formatted for you as dates.

Aims

The aims of this session are:

- Estimate cumulative incidence functions using non-parametric methods (the Aalen-Johansen estimator), and plot and interpret these.
- Compare cumulative incidence functions between groups.
- Understand why the Kaplan-Meier estimator is not in general suitable for estimating cumulative incidences when there are competing events.

Question 1

- Load the data.
- How many people died from any cause, and how many had an AAA-death and a non-AAA-death?
- Calculate the follow-up time (since randomisation) in years using `dateran` and `timeout`, and call this variable `time`. This will be the time variable that we use in subsequent analyses.

SOLUTION

```
#load the data
load(file="datasets/aaatrial.RData")

#number of deaths by type
dim(aaatrial)
```

```
[1] 13562    10
```

```
table(aaatrial$alldeath)
```

```
  0    1  
9522 4040
```

```
table(aaatrial$aaadeath)
```

```
  0    1  
13457  105
```

```
table(aaatrial$deathtype)
```

```
  0    1    2  
9522 3935  105
```

```
#creating time variable  
aaatrial$time<-as.numeric(aaatrial$timeout-aaatrial$dateran)/365.25
```

There are 4040 deaths. Of these, 105 are AAA-deaths and 3935 are deaths from other causes (non-AAA-deaths).

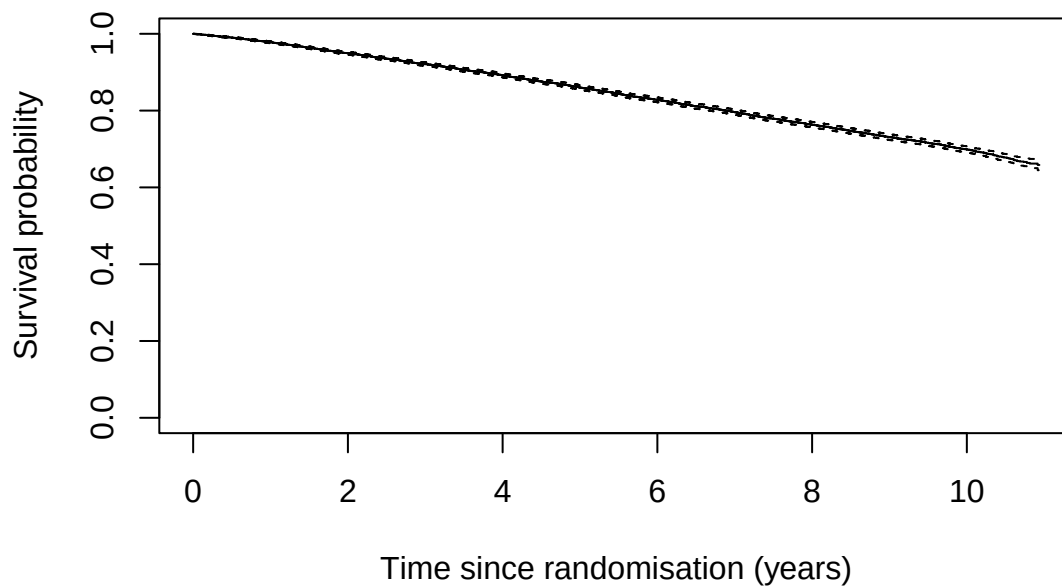
Question 2

We will begin by focusing on death from any cause using the `alldeath` variable.

- Plot the Kaplan-Meier survival curves for all-cause mortality.
- What is the risk of death from any cause up to 5 years?

SOLUTION

```
#Kaplan-Meier estimates of overall survival probabilities  
km <- survfit(Surv(time, alldeath)~1, data=aaatrial)  
  
#Plot Kaplan-Meier survival curve  
plot(km, ylim=c(0,1), xlab="Time since randomisation (years)",  
      ylab="Survival probability", conf.int=T)
```



```
#Obtain estimates at 1,5,10 years
summary(km, times=c(1,5,10))
```

```
Call: survfit(formula = Surv(time, alldeath) ~ 1, data = aaatrial)
```

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	13255	300	0.978	0.00126	0.975	0.980
5	11631	1597	0.860	0.00298	0.854	0.866
10	4269	2048	0.699	0.00408	0.691	0.707

The survival probability at 5 years is 0.86. The risk of death from any cause up to 5 years is therefore 14%.

Question 3

We will now move on to considering AAA-deaths and non-AAA-deaths as competing events. We first explore how to use the `Surv()` function in R when there are competing risks.

Use the code below and investigate the output. What information is given for (i) people who have an AAA-death, (ii) people who have a non-AAA-death, (iii) people who are censored?

```
Surv(time=aaatrial$time,event=as.factor(aaatrial$deathtype))
```

SOLUTION

```
# Looking at the output for the first 10 individuals
# Surv(time=aaatrial$time,event=as.factor(aaatrial$deathtype))
Surv(time=aaatrial$time,event=as.factor(aaatrial$deathtype))[1:10]
```

```
[1] 3.7864476:2 10.3216975+ 0.2190281:1 10.3216975+ 2.8774812:1
[6] 10.3216975+ 10.3216975+ 4.7720739:1 10.3216975+ 10.3216975+
```

- Using `Surv()` in this way records information on the person's event or censoring time, and whether the end of follow-up for the person is due to an event (:1 - non-AAA-death or :2 - AAA-death) or is due to censoring (+).
- The first person in the data died from an AAA-death at 3.78 years. The second person was censored at 10.32 years. The third person died from a non-AAA-death at 0.219 years.
- Note it is very important to have `deathtype` as a factor variable when using `Surv()` to analyse data with competing risks using the methods in this practical.

Question 4

- Estimate the cumulative incidences for AAA-deaths and non-AAA-deaths using

```
cinc <- survfit(Surv(time, as.factor(deathtype)) ~1, data=aaatrial)
```

- The resulting cumulative incidence curves can be plotted using

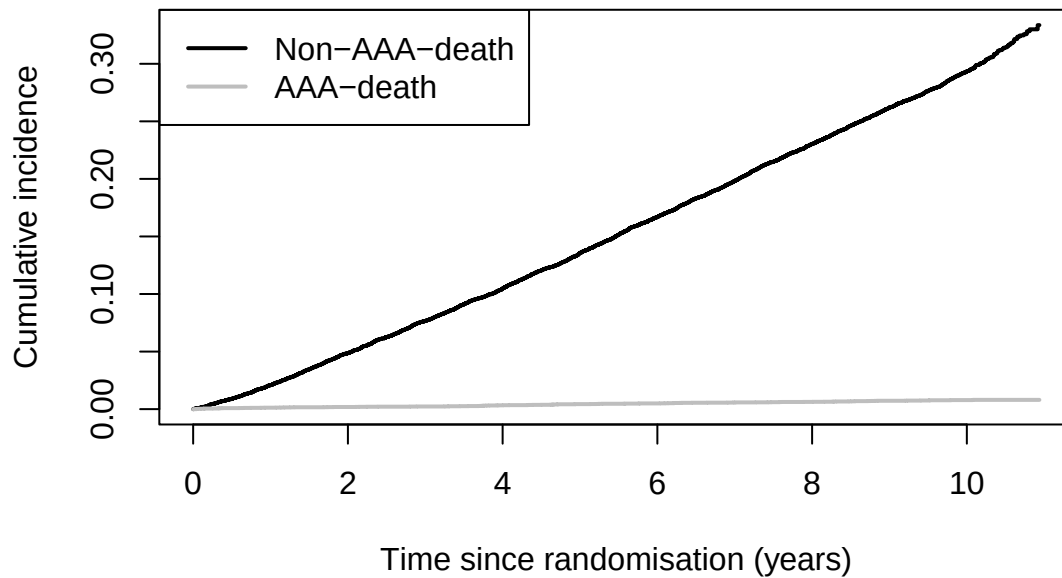
```
plot(cinc, lwd=2,col=c("black","grey"))
legend("topleft",c("Non-AAA-death","AAA-death"),
      lwd=2,col=c("black","grey"))
```

- What are the cumulative incidences for AAA-deaths and non-AAA-deaths up to 5 years?

SOLUTION

```
#Estimate the cumulative incidences using the Aalen-Johansen estimator
cinc <- survfit(Surv(time, as.factor(deathtype)) ~1, data=aaatrial)

#plot the cumulative incidences
plot(cinc, lwd=2,col=c("black","grey"),
     xlab="Time since randomisation (years)",
     ylab="Cumulative incidence")
legend("topleft",c("Non-AAA-death","AAA-death"),
     lwd=2,col=c("black","grey"))
```



```
#cumulative incidences at specified times
summary(cinc,times=c(1,5,10))
```

```
Call: survfit(formula = Surv(time, as.factor(deathtype)) ~ 1, data = aaatrial)
```

time	n.risk	n.event	Pr((s0))	Pr(1)	Pr(2)
1	13255	300	0.978	0.0209	0.00125
5	11631	1597	0.860	0.1357	0.00436
10	4269	2048	0.699	0.2934	0.00785

The cumulative incidence of AAA-death is very low, and much lower than the cumulative incidence of non-AAA-death. The estimated cumulative incidence of AAA-death up to 5 years is 0.44%. The estimated cumulative incidence of non-AAA-death up to 5 years is 13.6%.

Question 5

- Returning to considering death from any cause (`alldeath`), obtain Kaplan-Meier plots by treatment group.
- What is the risk of death from any cause up to 5 years in the two groups?
- Use a log-rank test to test the null hypothesis that there is no difference between the survival curves in the two groups.

SOLUTION

```
#Kaplan-Meier estimates of survival probabilities by group
km.group <- survfit(Surv(time, alldeath)~group, data=aaatrial)

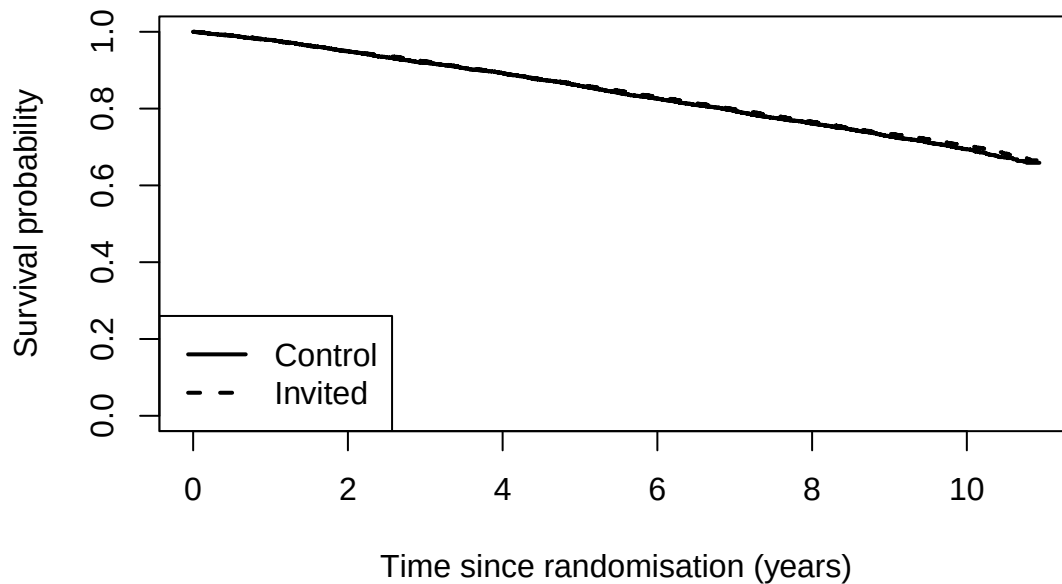
#Obtain estimates at 1,5,10 years
summary(km.group, times=c(1,5,10))
```

Call: survfit(formula = Surv(time, alldeath) ~ group, data = aaatrial)

group=0						
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	6665	144	0.979	0.00174	0.975	0.982
5	5835	815	0.859	0.00422	0.851	0.867
10	2081	1049	0.694	0.00581	0.683	0.705

group=1						
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	6590	156	0.977	0.00183	0.973	0.980
5	5796	782	0.861	0.00422	0.853	0.869
10	2188	999	0.704	0.00574	0.693	0.715

```
#Plot Kaplan-Meier survival curves
plot(km.group, ylim=c(0,1),
      lwd=2,lty=c(1,2),
      xlab="Time since randomisation (years)",
      ylab="Survival probability", conf.int=F)
legend("bottomleft",c("Control","Invited"),
      lwd=2,lty=c(1,2))
```



```
#log rank test
survdif(Surv(time, alldeath)~group, data=aaatrial)
```

Call:

```
survdif(formula = Surv(time, alldeath) ~ group, data = aaatrial)
```

	N	Observed	Expected	$(O-E)^2/E$	$(O-E)^2/V$
group=0	6815	2054	2021	0.55	1.1
group=1	6747	1986	2019	0.55	1.1

Chisq= 1.1 on 1 degrees of freedom, p= 0.3

- There is very little difference between the overall survival curves in the two treatment groups. The estimated survival probability at 5 years is 0.859 in the control group and 0.861 in the invited group.
- The log rank test has a p-value of 0.3. There is no evidence against the null hypothesis of no difference between the survival curves in the two groups.

Question 6

We now return to considering the competing events of AAA-death and non-AAA death.

- Estimate the cumulative incidence function for AAA-deaths and non-AAA-deaths in each treatment group using

```
cinc.group <- survfit(Surv(time, as.factor(deathtype)) ~ group, data=aaatrial)
```

- The resulting cumulative incidence curves can be plotted using

```
plot(cinc.group, lwd=2, lty=c(1,2,1,2), col=c("black", "black", "grey", "grey"),
     xlab="Time since randomisation (years)",
     ylab="Cumulative incidence")
legend("topleft", c("Control, Non-AAA-death", "Invited, Non-AAA-death",
                    "Control, AAA-death", "Invited, AAA-death"),
       lwd=2, lty=c(1,2,1,2), col=c("black", "black", "grey", "grey"))
```

- What are the cumulative incidences for AAA-deaths and non-AAA-deaths up to 5 years in the two groups?
- Perform Gray's test of the null hypothesis of no difference in the cumulative incidences for a given cause of death by group. This can be done using the `cuminc` function from the `cmprsk` package as follows:

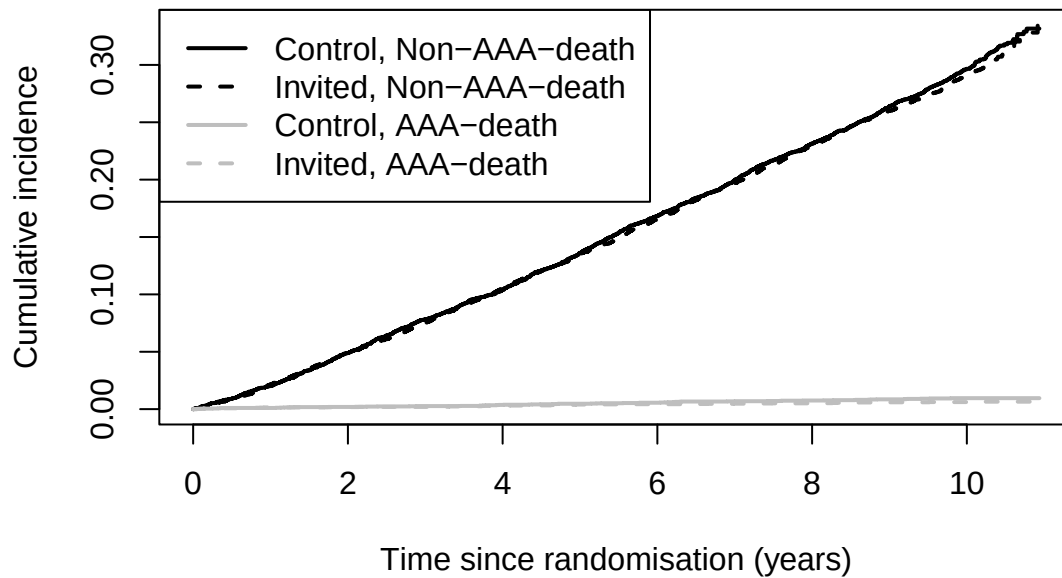
```
cuminc(ftime=aaatrial$time, fstatus=aaatrial$deathtype, group=aaatrial$group)
```

EXTRA: Check out what is stored in `cinc.group$time`, `cinc.group$strata` and `cinc.group$pstate` and see if you can use this information to make a plot of the cumulative incidences of AAA-death only in the two groups.

SOLUTION

```
#Estimate the cumulative incidences using the Aalen-Johansen estimator
cinc.group <- survfit(Surv(time, as.factor(deathtype)) ~ group, data=aaatrial)

#plot the cumulative incidences
plot(cinc.group,
     lwd=2, lty=c(1,2,1,2), col=c("black", "black", "grey", "grey"),
     xlab="Time since randomisation (years)",
     ylab="Cumulative incidence")
legend("topleft", c("Control, Non-AAA-death", "Invited, Non-AAA-death",
                    "Control, AAA-death", "Invited, AAA-death"),
       lwd=2, lty=c(1,2,1,2), col=c("black", "black", "grey", "grey"))
```



```
#cumulative incidences at specified times
summary(cinc.group, times=c(1, 5, 10))
```

Call: `survfit(formula = Surv(time, as.factor(deathtype)) ~ group, data = aaatrial)`

```
group=0
time n.risk n.event Pr((s0)) Pr(1) Pr(2)
1    6665    144    0.979 0.0201 0.00103
5    5835    815    0.859 0.1362 0.00470
10   2081   1049    0.694 0.2966 0.00955
```

```
group=1
time n.risk n.event Pr((s0)) Pr(1) Pr(2)
1    6590    156    0.977 0.0216 0.00148
5    5796    782    0.861 0.1351 0.00401
10   2188    999    0.704 0.2902 0.00614
```

```
#Performing Gray's test using
library(cmprsk)

cuminc(ftime=aaatrial$time, fstatus=aaatrial$deathtype, group=aaatrial$group)
```

Tests:

stat	pv	df
------	----	----

```

1 0.4591326 0.49802963 1
2 4.0832244 0.04331093 1
Estimates and Variances:
$est
      2      4      6      8     10
0 1 0.048902374 0.104019997 0.168342948 0.231455119 0.296580095
1 1 0.048328207 0.105598078 0.165442251 0.229045718 0.290189218
0 2 0.001908650 0.003672580 0.005586004 0.007502317 0.009548797
1 2 0.001778726 0.002965781 0.004302079 0.005045082 0.006139720

$var
      2      4      6      8     10
0 1 6.831427e-06 1.369562e-05 2.058956e-05 2.618531e-05 3.312115e-05
1 1 6.819290e-06 1.401024e-05 2.049170e-05 2.622050e-05 3.256366e-05
0 2 2.797335e-07 5.376166e-07 8.166906e-07 1.095535e-06 1.452445e-06
1 2 2.632261e-07 4.385571e-07 6.355605e-07 7.449609e-07 9.166547e-07

```

- The estimated cumulative incidence of AAA-death up to 5 years is 0.96% in the control group and 0.61% in the invited group. The estimated cumulative incidence of non-AAA-death up to 5 years is 29.7% in the control group and 29.0% in the invited group.
- Gray's test gives a p-value (`pv` in the output) of 0.498 for non-AAA-deaths, indicating no evidence against the null hypothesis of no difference between the cumulative incidences in the two groups.
- Gray's test gives a p-value of 0.043 for AAA-deaths, indicating some evidence against the null hypothesis of no difference between the cumulative incidences in the two groups.

In `cinc.group$time` we have the (unique) times at which events and censorings occur. The first 1666 elements of `cinc.group$time` refer to the control group and the remaining 1623 elements of `cinc.group$time` refer to the invited group. `cinc.group$strata` tells us how many unique event and censoring times there are in the two groups. You can check this using:

```
length(unique(aaatrial$time[aaatrial$group==0]))
```

```
[1] 1666
```

```
length(unique(aaatrial$time[aaatrial$group==1]))
```

```
[1] 1623
```

- `cinc.group$pstate` has 3289 rows (1666+1623=3289) and 3 columns. `pstate` stands for the probability of being in a given state.

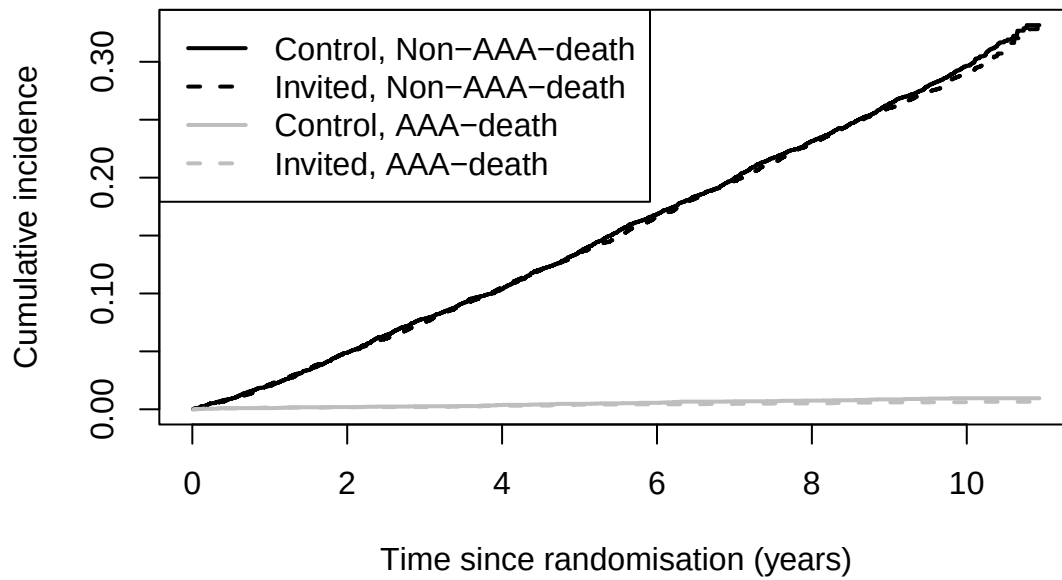
- Column 1 (`cinc.group$pstate[,1]`) shows the cumulative incidence of censoring at each time, with the first 1666 rows being for the control group and the remaining 1623 rows being for the invited group.
- Column 2 (`cinc.group$pstate[,2]`) shows the cumulative incidence of the event type `deathtype=1` (non-AAA-death) at each time, with the first 1666 rows being for the control group and the remaining 1623 rows being for the invited group.
- Column 3 (`cinc.group$pstate[,3]`) shows the cumulative incidence of the event type `deathtype=2` (AAA-death) at each time, with the first 1666 rows being for the control group and the remaining 1623 rows being for the invited group.

```
#Another way of creating the plot that gives more control over its appearance
cinc.group$strata
```

```
group=0 group=1
1666    1623
```

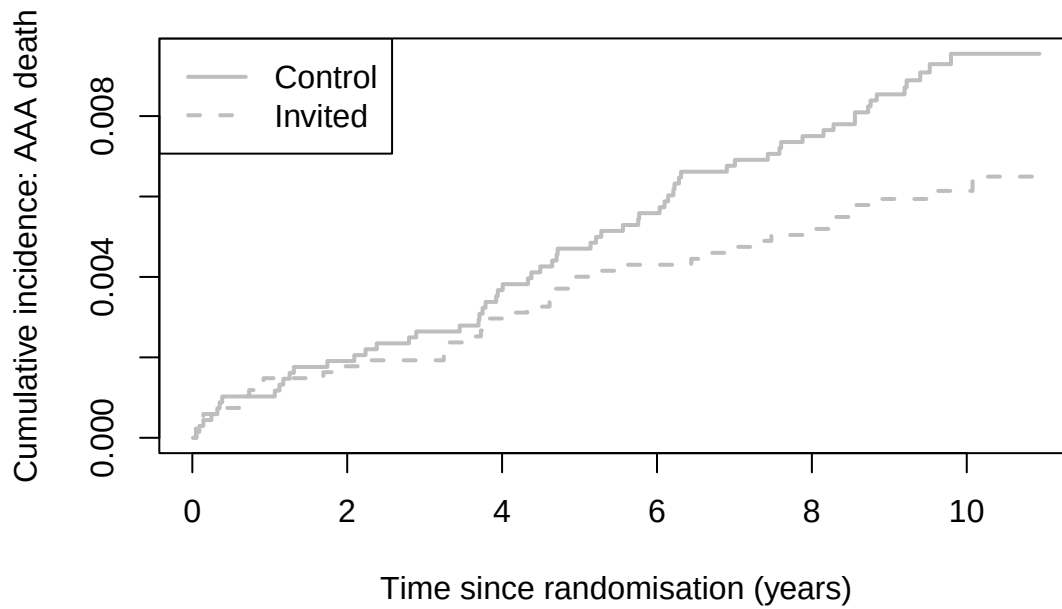
```
time.group0<-cinc.group$time[1:cinc.group$strata[1]]
time.group1<-cinc.group$time[(cinc.group$strata[1]+1):
                             (cinc.group$strata[1]+cinc.group$strata[2])]
cinc.group0<-cinc.group$pstate[1:cinc.group$strata[1],]
cinc.group1<-cinc.group$pstate[(cinc.group$strata[1]+1):
                             (cinc.group$strata[1]+cinc.group$strata[2]),]
plot(time.group0,cinc.group0[,2],
     type="s",lwd=2,lty=1,col="black",
     xlab="Time since randomisation (years)",
     ylab="Cumulative incidence")#group 0, non-AAA-deaths
points(time.group0,cinc.group0[,3],
       type="s",lwd=2,lty=1,col="grey")#group 0, AAA-deaths
points(time.group1,cinc.group1[,2],
       type="s",lwd=2,lty=2,col="black")#group 1, non-AAA-deaths
points(time.group1,cinc.group1[,3],
       type="s",lwd=2,lty=2,col="grey")#group 1, AAA-deaths

legend("topleft",c("Control, Non-AAA-death","Invited, Non-AAA-death",
                  "Control, AAA-death","Invited, AAA-death"),
      lwd=2,lty=c(1,2,1,2),col=c("black","black","grey","grey"))
```



```
#zooming in on the cumulative incidences for AAA-deaths
plot(time.group0,cinc.group0[,3],
     type="s",lwd=2,lty=1,col="grey",
     xlab="Time since randomisation (years)",
     ylab="Cumulative incidence: AAA death")#group 0, AAA-deaths
points(time.group1,cinc.group1[,3],
       type="s",lwd=2,lty=2,col="grey")#group 1, AAA-deaths

legend("topleft",c("Control","Invited"),
      lwd=2,lty=c(1,2),col=c("grey","grey"))
```



Question 7

- Use `Surv(time=aaatrial$time,event=aaatrial$aaadeath)` and investigate the output. What information is given for (i) people who have an AAA-death, (ii) people who have a non-AAA-death, (iii) people who are censored? Compare this with what you obtained in Question 3 where you looked at `Surv(time=aaatrial$time,event=as.factor(aaatrial$deathtype))`.
- Use the Kaplan-Meier estimator to estimate the cumulative incidence of AAA-death using

```
cinc.km <- survfit(Surv(time, aaadeath) ~1, data=aaatrial)
```

- Plot the cumulative incidence function based on `cinc.km` and compare this with what you obtained in `cinc` (i.e. based on the Aalen-Johansen estimator). Compare the estimates and comment on why the results are different.

SOLUTION

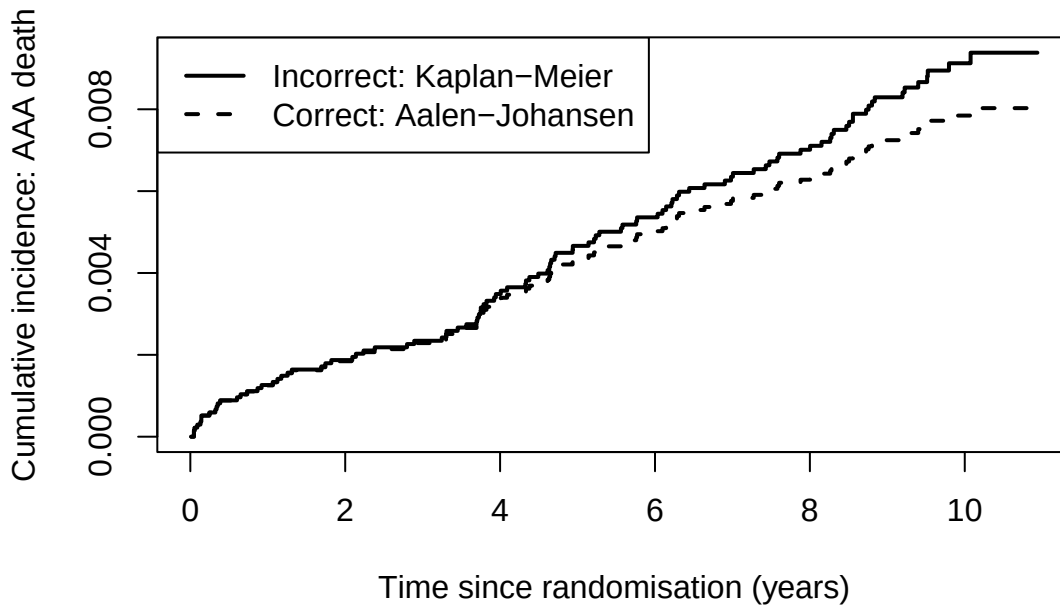
```
Surv(time=aaatrial$time,event=aaatrial$aaadeath)[1:10]
```

```
[1] 3.7864476 10.3216975+ 0.2190281+ 10.3216975+ 2.8774812+ 10.3216975+
[7] 10.3216975+ 4.7720739+ 10.3216975+ 10.3216975+
```

```
Surv(time=aaatrial$time,event=as.factor(aaatrial$deathtype))[1:10]
```

```
[1] 3.7864476:2 10.3216975+ 0.2190281:1 10.3216975+ 2.8774812:1
[6] 10.3216975+ 10.3216975+ 4.7720739:1 10.3216975+ 10.3216975+
```

```
cinc.km <- survfit(Surv(time, aaadeath) ~1, data=aaatrial)
plot(cinc.km$time,1-cinc.km$surv,
     type="s",lwd=2,
     xlab="Time since randomisation (years)",
     ylab="Cumulative incidence: AAA death")
lines(cinc$time,cinc$pstate[,3],
      type="s",lwd=2,lty=2)
legend("topleft",
      c("Incorrect: Kaplan-Meier","Correct: Aalen-Johansen"),
      lwd=2,lty=c(1,2))
```



As expected from theory, the Kaplan-Meier estimator overestimates the cumulative incidence of AAA death, as compared to the estimate obtained using the Aalen-Johansen estimator. This is because the Kaplan-Meier estimator treats non-AAA-deaths in the same way that it treats censoring, resulting in an implicit assumption that individuals who have had a non-AAA-death could have an AAA-death after their time of having a non-AAA-death, which does not make sense.